Mechanical Properties

Wednesday, October 30, 2024 12:46 AM

Stress (σ)

Force / Area

Reaction to applied force

$\frac{N}{M^2} = Pa$; $MPa = \frac{MN}{m^2} = \frac{N}{mm^2}$

$\frac{Kg}{cm^2}$; $\frac{lb}{in^2}$

Strain (ε)

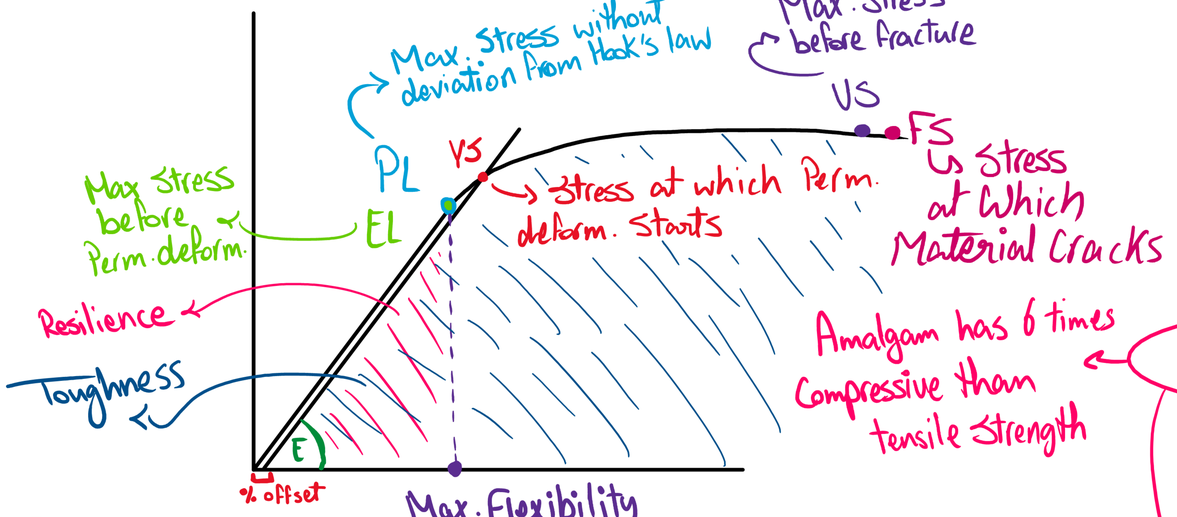
$\frac{\Delta L}{L_o}$

Change in length per unit length

Elastic → Temp

Plastic → Perm.

Poisson's Ratio: $\mu = \frac{\text{Lateral st.}}{\text{axial st.}} = 0.3$ usually



Impression
- Ortho. Wires
- Strain when stress = PL → $\frac{PL}{E}$

Brittleness

Material Fracture Near PL

Fracture by Crack Propagation

Amalgam has 6 times Compressive than tensile strength

Tensile strength of brittle materials is measured by Indirect Tensile Test

$\sigma_t = \frac{2 \times load}{\pi \times D \times T}$

Modulus of Shear (G)

$\frac{E}{2(1+\mu)}$ → 46% of E

E under shear stresses

Modulus of Elasticity (E)

$\frac{stress}{strain}$

Constant of Proportionality bet. stress & strain (slope)

Doesn't Change

Base Metal > Gold

Metals > Plastics

Rigidity Vs Flexibility

E ↑ → Even stress distribution → Used in thin Section Without fear of bending
- Long span Bridge
- Denture Base

Base (Cement) → to support brittle Restorative Material

Ability to be deformed Plastically Under

- Tensile → Ductility
- Compressive → Malleability

Tensile Strength of Ductile Materials is Measured be % Elongation = $\frac{\Delta L}{L_o} \times 100$

↳ Gold = 20%

indicates Workability on burnishing

Resilience

Energy absorbed When stressed to PL

$\frac{1}{2} \times Stress \times Strain$

$\frac{mMN}{m^3}$

Load Removed ⇒ Energy Released

Ortho. Wires → Released Energy Causes Movement of teeth

Denture Base → To avoid transmitting Forces to bone.

Toughness

Energy absorbed When stressed to FS

High PL + High Ductility = High Toughness

With defects → Fracture Toughness